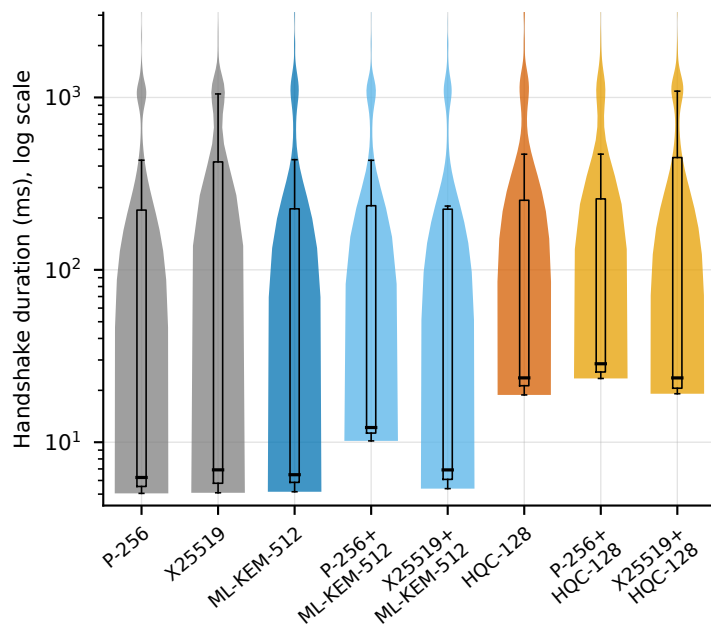
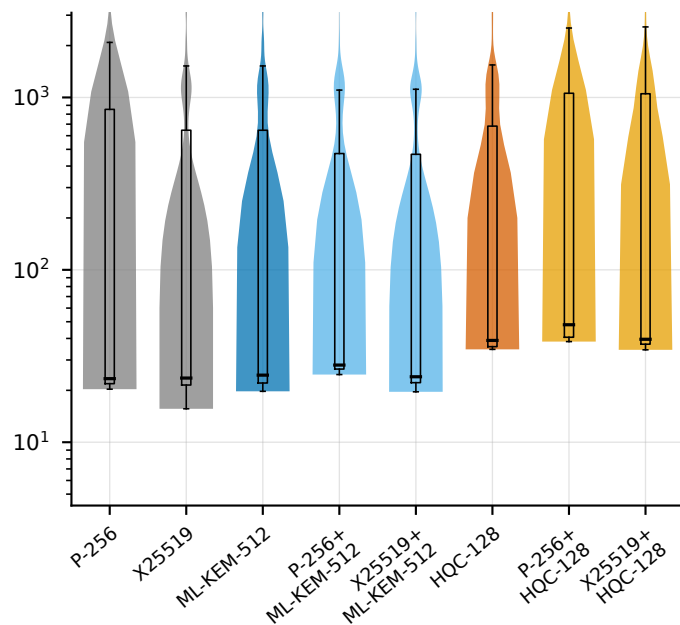


(a)

ed25519 (classical)

**(b)**

mldsa44 (post-quantum)

**(c)**
 Classical KEX
 ML-KEM (pure)
 ML-KEM (hybrid)
 HQC (pure)
 HQC (hybrid)

KEM	ed25519 Mean (ms)	ed25519 CV (%)	ed25519 Max (ms)	ed25519 Fail	mldsa44 Mean (ms)	mldsa44 CV (%)	mldsa44 Max (ms)	mldsa44 Fail
P-256	267.9	195.3	4.08k	0	595.4	472.2	52.54k	0
X25519	339.6	242.5	13.15k	0	406.1	176.8	4.40k	0
ML-KEM-512	304.3	214.6	6.37k	0	458.4	203.3	11.45k	0
P-256+ML-KEM-512	279.6	218.4	7.20k	0	396.9	204.4	8.49k	0
X25519+ML-KEM-512	275.7	209.6	4.55k	0	377.8	179.6	4.17k	0
HQC-128	390.9	195.9	6.60k	0	485.5	220.0	16.42k	0
P-256+HQC-128	345.0	225.1	10.77k	0	576.6	433.3	52.84k	1/500
X25519+HQC-128	361.8	165.5	5.17k	0	541.0	271.4	27.63k	0